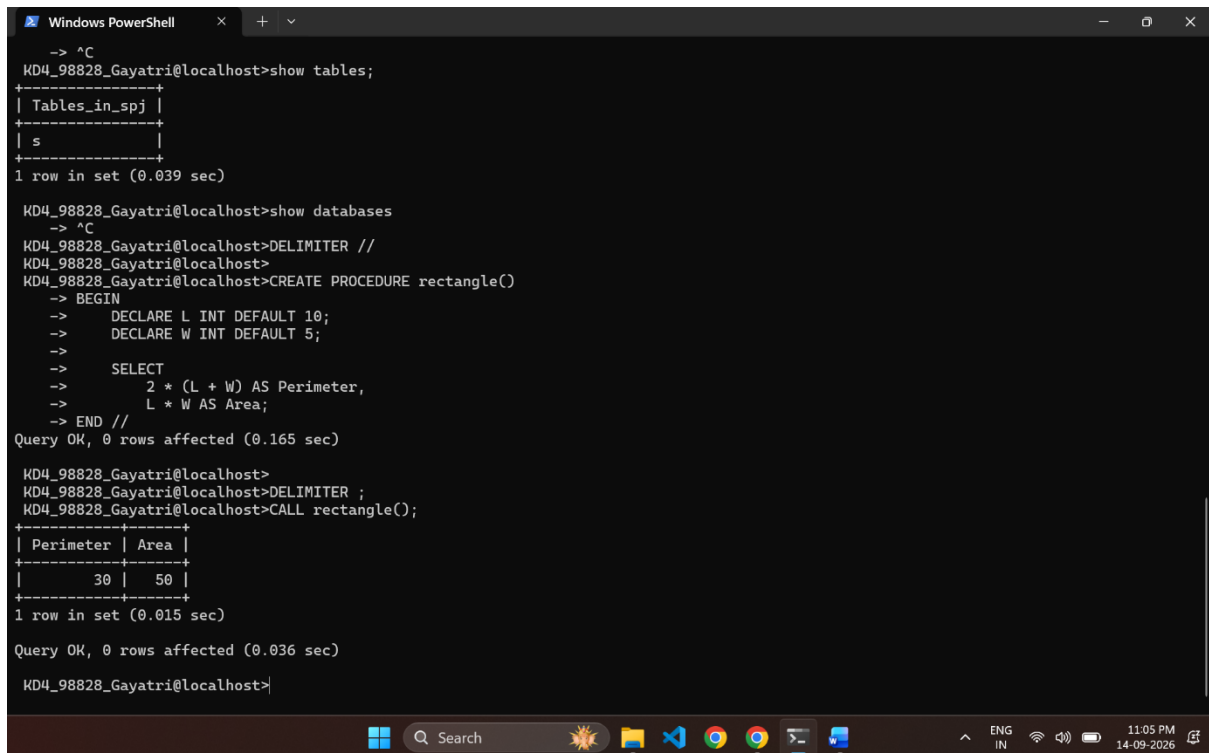


ASSIGNMENT=10

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).



```
Windows PowerShell
-> ^C
KD4_98828_Gayatri@localhost>show tables;
+-----+
| Tables_in_spj |
+-----+
| s              |
+-----+
1 row in set (0.039 sec)

KD4_98828_Gayatri@localhost>show databases
-> ^C
KD4_98828_Gayatri@localhost>DELIMITER //
KD4_98828_Gayatri@localhost>CREATE PROCEDURE rectangle()
-> BEGIN
->   DECLARE L INT DEFAULT 10;
->   DECLARE W INT DEFAULT 5;
->
->   SELECT
->       2 * (L + W) AS Perimeter,
->       L * W AS Area;
-> END //
Query OK, 0 rows affected (0.165 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL rectangle();
+-----+-----+
| Perimeter | Area |
+-----+-----+
|         30 |    50 |
+-----+-----+
1 row in set (0.015 sec)

Query OK, 0 rows affected (0.036 sec)

KD4_98828_Gayatri@localhost>
```

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
Windows PowerShell
--> );
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'cube INT' at line 4
KD4_98828_Gayatri@localhost>CREATE TABLE temp (
-->   num INT,
-->   square_value INT,
-->   cube_value INT
--> );
Query OK, 0 rows affected (0.283 sec)

KD4_98828_Gayatri@localhost>DELIMITER //
KD4_98828_Gayatri@localhost>CREATE PROCEDURE num_calculation()
--> BEGIN
-->   DECLARE num INT DEFAULT 5;
-->
-->   INSERT INTO temp
-->   VALUES (num, num * num, num * num * num);
--> END //
Query OK, 0 rows affected (0.095 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL num_calculation();
Query OK, 1 row affected (0.090 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM temp;
+-----+-----+-----+
| num | square_value | cube_value |
+-----+-----+-----+
| 5 | 25 | 125 |
+-----+-----+-----+
1 row in set (0.009 sec)

KD4_98828_Gayatri@localhost>
```

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
Windows PowerShell
KD4_98828_Gayatri@localhost>CALL num_calculation();
Query OK, 1 row affected (0.090 sec)

KD4_98828_Gayatri@localhost>SELECT * FROM temp;
+-----+-----+-----+
| num | square_value | cube_value |
+-----+-----+-----+
| 5 | 25 | 125 |
+-----+-----+-----+
1 row in set (0.009 sec)

KD4_98828_Gayatri@localhost>DELIMITER //
KD4_98828_Gayatri@localhost>CREATE PROCEDURE f_to_c(IN F DOUBLE)
--> BEGIN
-->   DECLARE C DOUBLE;
-->
-->   SET C = (F - 32) * 5 / 9;
-->
-->   SELECT F AS Fahrenheit, C AS Celsius;
--> END //
Query OK, 0 rows affected (0.092 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL f_to_c(100);
+-----+-----+
| Fahrenheit | Celsius |
+-----+-----+
| 100 | 37.777777777778 |
+-----+-----+
1 row in set (0.060 sec)

Query OK, 0 rows affected (0.101 sec)

KD4_98828_Gayatri@localhost>
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```
Windows PowerShell
+-----+
1 row in set (0.060 sec)

Query OK, 0 rows affected (0.101 sec)

KD4_98828_Gayatri@localhost>DELIMITER //
KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>CREATE PROCEDURE convert_inches(IN total_inches INT)
-> BEGIN
->   DECLARE yards INT;
->   DECLARE feet INT;
->   DECLARE inches INT;
->
->   SET yards = total_inches DIV 36;
->   SET feet = (total_inches MOD 36) DIV 12;
->   SET inches = total_inches MOD 12;
->
->   SELECT yards AS Yards,
->          feet AS Feet,
->          inches AS Inches;
-> END //
Query OK, 0 rows affected (0.101 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL convert_inches(124);
+-----+
| Yards | Feet | Inches |
+-----+
| 3 | 1 | 4 |
+-----+
1 row in set (0.011 sec)

Query OK, 0 rows affected (0.024 sec)

KD4_98828_Gayatri@localhost>
```

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100

```
Windows PowerShell
KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>CREATE PROCEDURE check_product(IN num1 DOUBLE, IN num2 DOUBLE)
-> BEGIN
->   DECLARE product DOUBLE;
->
->   SET product = num1 * num2;
->
->   IF product >= 100 THEN
->     SELECT product AS Product,
->            'Product is equal to or greater than 100' AS Result;
->   ELSE
->     SELECT product AS Product,
->            'Product is less than 100' AS Result;
->   END IF;
-> END //
Query OK, 0 rows affected (0.099 sec)

KD4_98828_Gayatri@localhost>
KD4_98828_Gayatri@localhost>DELIMITER ;
KD4_98828_Gayatri@localhost>CALL check_product(20, 5);
+-----+
| Product | Result
+-----+
| 100 | Product is equal to or greater than 100 |
+-----+
1 row in set (0.011 sec)

Query OK, 0 rows affected (0.042 sec)

KD4_98828_Gayatri@localhost>
```